

DIVYA INDRA KUMAR

Data Scientist

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Summary

Data Scientist with 4.5+ years of experience in data modelling and algorithm development for enterprise solutions. Expert in statistical inference, computational analysis, and experimental design methodologies. Delivers substantial cost reductions through automated intelligence frameworks, performance optimization strategies, and cross-functional collaboration, consistently enhancing operational efficiency across organizational platforms.

Skills

Methodologies: SDLC, Agile (Scrum, Kanban), Waterfall

Programming Languages: Python, R, SQL

Libraries: Pandas, NumPy, SciPy, XGBoost, LightGBM, Scikit Learn, TensorFlow, PyTorch, Keras

Databases and Data Processing tools: MySQL, PostgreSQL, MongoDB, PySpark, Apache Airflow, Talend, Alteryx

Data Visualization: Tableau, Power BI, Excel (Advanced)

Cloud Platforms: Microsoft Azure (Data Lake, Data Factory, Azure DevOps), AWS (EC2, S3, RedShift, SageMaker)

Statistical Analysis: A/B Testing, Hypothesis Testing, Bayesian Methods, Predictive Analysis, Time Series Analysis, ANOVA, Regression Analysis

Machine Learning Algorithms: Supervised and Unsupervised Learning, Random Forests, Clustering Models, Decision Trees, Linear and Logistic Regression, SVM, NLP, Large Language Models (LLM), Feature Engineering, Deep learning, Forecasting

Version Control & Containerization: Git, GitHub, Docker, Kubernetes

Soft Skills: Critical thinking, Problem solving, Attention to detail, Strong written and verbal communication skills, Data storytelling, Collaboration with cross-functional team, Stakeholder management

Education

Bachelor of Engineering | Visvesvaraya Technological University, India

July 2010

Work Experience

Data Scientist | Cardinal Health, North Carolina

January 2023 - Current

- Collected, transformed, and explored 500K+ anonymized patient records using **SQL**, **PySpark**, and **Azure Data Factory** to build robust data pipelines, developing deep expertise in clinical datasets.
- Designed **machine learning models** using **Python**, **TensorFlow**, and **Scikit-learn** for predictive analytics, achieving accuracy in patient outcome forecasting while reducing hospital readmission rates by 18%.
- Implemented retrieval augmented generation pipeline using **LangChain** and **vector databases**, developing LLM-powered clinical decision support system that improved diagnostic accuracy and reduced treatment delays by 25%.
- Built end-to-end **MLOps framework** with **Docker**, **Azure DevOps**, and continuous integration practices, enabling automated model deployment and monitoring across 2 regional healthcare facilities.
- Conducted rigorous **A/B testing** and **statistical analysis** using Bayesian methods and hypothesis testing, validating model performance and ensuring \$450K annual cost savings through optimized resource allocation.
- Coordinated between engineering teams and product analysts while creating interactive **Power BI** dashboards, translating complex AI insights into actionable business intelligence for 25+ clinical stakeholders.
- Delivered comprehensive reports and presentations to department managers, demonstrating ROI through **time series analysis** and **predictive modelling** that supported tactical healthcare technology adoption decisions.

Data Scientist | PNC, North Carolina

March 2021 - December 2022

- Built automated data pipeline using **Python**, **SQL**, and **AWS Glue** to ingest customer data from **Snowflake** data warehouse and REST APIs, implementing data quality validation that improved downstream analytics accuracy by 35%.
- Collaborated with marketing and compliance teams to understand regulatory requirements and translate customer retention questions into analytical frameworks, conducting exploratory data analysis using **Pandas**, **NumPy**, and **A/B testing**.
- Developed machine learning models using **Scikit-Learn**, **XGBoost**, and **AWS SageMaker** for AI-powered churn prediction, achieving accuracy through **feature engineering** while ensuring **data governance** and model interpretability standards.
- Deployed **predictive models** using **AWS SageMaker** pipelines with **Git version control**, implementing automated model monitoring, performance tracking, and retraining workflows that maintained prediction accuracy above 85%.
- Created **Tableau** dashboards to visualize customer segmentation insights and model predictions with proper data lineage documentation, enabling stakeholders to make data-driven decisions improving campaign effectiveness by 25%.
- Enhanced the analytics system with **time series forecasting** using **regression techniques** and **statistical methods** including **ANOVA**, supporting quarterly business planning with forecast accuracy while maintaining audit trail requirements.
- Optimized model training workflows using **AWS EC2** compute resources and automated scheduling, reducing processing time while collaborating with data engineering teams to maintain secure, scalable ML infrastructure with proper access controls.

Certifications

Professional Scrum Master (PSM)

Java Certificate from Wake Tech University, Raleigh NC